

Research paper

Hybrid CNN-LSTM model for urban energy load forecasting with IGA-XAI for smart grids: Peak and off-peak variability insights

Sarwar Morshed Shawon^a, Shah Nawaz Haider^b, Arnab Barua^a, Steve Austin^b,
Ifaz Ahmed Adan^a, Mohammad Shahadat Hossain^a, H.T. Zubair^{c,d,*}

^a Department of Electrical and Electronic Engineering, University of Science and Technology Chittagong, Chattogram, 4202, Bangladesh

^b Department of Computer Science and Engineering, University of Science and Technology Chittagong, Chattogram, 4202, Bangladesh

^c Faculty of Computing and Informatics (FCI), Multimedia University, Cyberjaya, 63100, Malaysia

^d Centre for Fibre Networking and Communication, COE for Intelligent Network, Multimedia University, 63100 Cyberjaya, Malaysia

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ABSTRACT

Urban areas face time-specific stochastic energy demand pattern, complicating the distribution of energy efficiently, keeping costs in sight. Present approaches seldom consider the intricate and multi-scaling dynamics of urban energy requirements specially during the Peak and Off-peak periods, resulting in inefficiencies and potential overloading of energy systems. To solve this, this study proposes a hybrid 1D-CNN-LSTM model to forecast short to medium-term electrical load across three critical daily interval such as Day Peak, Evening Peak and Off-Peak in urban context. Unlike most state-of-the-art models, limited to 24–72 h, the proposed model achieves class leading accuracy in day-ahead peak and off-peak load forecasts while maintaining robust performance for up to 7 days in advance. The proposed 1D-CNN-LSTM hybrid model, enabled by Integrated Gradients Attribution (IGA) based eXplainable AI (XAI) was capable of accurately forecasting the intricacies in the load patterns achieving class leading performance with R^2 of 0.9602, 0.948 and 0.9464 for Day Peak, Evening Peak and Off-Peak, respectively, for the city of Chattogram. The model was also tested for two other major metropolitan cities (Dhaka and Sylhet), proving its generalization capability. To further test robustness, Gaussian noise (5–20 %) was added to the meteorological inputs, and the CNN-LSTM model demonstrated resilience with only a modest 2.5 % average accuracy drop across all three peak demand periods. The proposed method enhances urban energy management by enabling more precise load forecasting, supporting smarter grid operations, improved reliability and progress towards sustainable energy systems.

1. Introduction

The significance of electricity has just kept increasing since the invention of the first practical electric bulb in 1879 by Thomas Alba Edison [1]. This surged greatly upon Edison's construction of Pearl Street Generating Station, the first permanent and commercial power station, in 1882 [2]. The demand was only found and predicted to rise as the weather conditions worsen [3]. Developing and developed countries in particular, are producing more electricity than ever to fight the

changing climate and rise of temperature, worldwide [4]. This further adds to the emission of carbon-based molecules to the atmosphere, further worsening the situation [5]. To maintain a balance between the demand and production of electricity, demand load forecasting and responsive energy generation plays a crucial role in both supplying the consumers with constant flow of electricity and doing so in a reasonable cost [6]. This method of production of energy, following a demand forecast, can impact carbon emissions and thus, help reach carbon-neutrality in the foreseeable future [7].

Abbreviations: : AFC-ANN, Adaptive Feature Combination - Artificial Neural Network; AI, Artificial Intelligence; ANN, Artificial Neural Network; ARIMA, Auto-Regressive Integrated Moving Average; CEEMDAN, Complete Ensemble Empirical Mode Decomposition with Adaptive Noise; CNN, Convolutional Neural Network; DNN, Deep Neural Network; DT, Decision Tree; DL, Deep Learning; FL, Federated Learning; GRU, Gated Recurrent Unit; GWDO, Group-Wise Differential Optimization; IES, Integrated Energy System; IGA, Integrated Gradients Attribution; IGWO, Improved Grey Wolf Optimizer; ISO-NE, Independent System Operator – New England; LDA, Linear Discriminant Analysis; LR, Logistic Regression; LSTM, Long Short-Term Memory; MAE, Mean Absolute Error.

* Corresponding author.

E-mail address: zubair@mmu.edu.my (H.T. Zubair).

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Bangladesh aims to generate 40 MW and 60 MW by 2030 and 2041 respectively [8]. But this is considered to be a highly emission-intensive plan. Though the access to power has increased by 285 % from 2008–2009 to 2020–2021, the country still lacks enough sustainability directed measures [9]. Challenges still persist in developing lightweight energy generation and distribution monitoring tools as per SDG 7 [10]. To combat this deficit, power demand forecasting or load forecasting can provide a strong platform to build a sustainable energy management system [11]. This is instrumental in enabling the nation with proper tools to achieve the SDG 7 by the United Nations, which puts sustainable growth and development at the very top of agendas [12]. Eras-Almeida and Egido-Aguilera, goes further in specifying the necessary actions to sustain SDG 7 in vulnerable circumstances [13].

The Chattogram city was selected as the case study region due to its geographical and socio-economic diversity. It includes dense urban centers, rural landscapes, coastal and riverine areas, as well as hilly regions, making it a miniature of varied energy consumption behaviors. This diversity supports the broader applicability of the proposed model to other metropolitan and mixed-use regions.

The focus of this study is improving load forecasting accuracy in the Chattogram metropolitan region through increasing the precision of demand forecast which aims to resolve the electricity consumption demand discrepancies. The research attempts to improve the reliability of forecasting by considering changes in electricity consumption during critical periods of the day with significant usage: daytime peak (from 9 am to 5 pm), evening peak (from 6 pm to 9 pm) and night-time off-peak (from 10 pm to 8 am). The study's purpose here is to enable better allocation of energy resources, curb forecasting inaccuracies and help optimize grid control in the region. This distinction between the peak and off-peak hours serves as crucial indicators to the load variability in overall energy demand [14].

As per the census report of Bangladesh Bureau of Statistics, the population of Chattogram city was 34,178,612 in 2022, which is 20.13 % of Bangladesh's total population [15]. Such classification of data requires the study to discover the more granulated usage habits that are significant in demand forecasting and eliminating power wastage as well. Furthermore, the time-series data on power consumption obtained from the study can be aggregated into relevant intervals during the day to strategically respond to where and when electricity load segregation and management is required. Such analytics can be built on a smart grid defined as a modernized power system that utilizes information and communication technologies to gather and act on information in an automated fashion to improve the reliability, security and efficiency of the electricity networks [16]. The purpose of this finding is to assist in valuing the tangible benefits of power quality improvement, grid reliability and increase support for regional energy efficiency initiatives. In doing so, this research also contributes to the planning of power forecasting model by addressing the fact of energy consumption in the city considering the implicit features embedded within an urban model. In addition, this study intends to better explore the impact of time-dependent features in load prediction problems, as lag values are important in any predictive task, particularly in time series forecasting problems, because of the necessity of capturing sequential dependencies and repeating patterns in data over time [17]. With utilizing past time steps, the models can learn how the introduction of historical data points affect the future predicted values. It also aims to examine and validate the generalizability of the model by applying best model architecture on load data from other cities.

With the very goal to contribute and attain more efficient energy production, distribution and consumption, this study aims to implement multiple DL approaches to accurately forecast daily load of the Chattogram city, sourced from the PGCB. The progress of ML has seen the emergence of new sophisticated techniques that capture complex patterns in data and enhance forecasting capabilities. The algorithms nominated for this study are- LSTM, GRU and CNN. All of which seem to have good performance upon studying recent relevant literature [18].

Besides, LSTM with attention was also tested in this study [19].

The hybrid CNN-LSTM model is well-suited for energy load forecasting as it combines LSTM's ability to model long-term temporal dependencies [20] with CNN's strength in extracting local patterns and fluctuations [21]. This synergy enables the model to effectively capture both daily trends and sharp variations in energy consumption, making it ideal for time-segmented forecasting tasks [22].

Although hybrid CNN-LSTM models have been used in energy forecasting, this study presents an enhanced version designed to capture Day Peak, Evening Peak and Off-Peak variations in metropolitan load patterns. By incorporating period-specific segmentation and optimized feature extraction, the model achieves improved accuracy and generalization over existing approaches. The peak and off-peak periods are the most crucial periods of the day, in terms of power management [23].

The study also utilizes eXplainable Artificial Intelligence (XAI) to interpret the DL models' predictions across the Day Peak, Evening Peak and Off-Peak hours. Integrated Gradients Attribution (IGA) is applied to identify which input features, such as- temperature, day type, or past consumption, contribute most towards the forecasts. This helps uncover the temporal importance and relevance of each feature, offering transparency and trust in the model's decision-making and ensuring its replicability.

The main objective of this study is to develop an accurate and interpretable novel hybrid model for short to medium-term load forecasting in urban settings. In order to do so, a hyper-tuned improved hybrid 1D-CNN-LSTM architecture is proposed to capture both spatial feature relationships and temporal patterns in multivariate time-series data. The model's performance is evaluated across Day Peak, Evening Peak and Off-Peak periods, with an optimal lag window of 7 identified through lag optimization. To enhance model transparency, IGA is employed for feature attribution, enabling XAI driven insights. The proposed approach is benchmarked against some state-of-the-art DL models, demonstrating superior accuracy and interpretability for real-life smart grid energy management applications.

2. Related works

Throughout the years and especially in the quest of a durable and reliable prediction model, a series of studies have been performed using both classical and also deep learning techniques. Key contributions are summarised in the section, indicating how the field started from statistical baselines and has been moving towards hybrid deep learning, attention-based models and XAI in a scalable way to make the technology more practical for the power grids.

Tiwari et al. [24] compared classical ML techniques including LR, DeT, RF, LDA, QDA, NB and SVM on Smart Grid Stability data (Kaggle) and found that SVM outperformed and proved the power of ML in structured categorization. By the same token, Khoshrou and Pauwels [25] compared the TRNSYS (Transient System Simulation Tool) against the MLR for assessing the energy performance of buildings and concluded that the MLR model is highly reliable ($R^2 > 0.9$, MAE and RMSE < 10 kWh/m²). Broader to load forecasting, Sayed et al. [26] also analysed MLR, RF, ANN and ARIMA with Dubai electricity consumption and found that although, at the district level, ARIMA could reach 93 % accuracy, ANN and RF achieved a better performance (about 97 % accuracy) and with much lower MAPE value (0.31 and 0.157), thus declaring them as the best methods for large scale system planning under load forecasting.

Drawing inspiration from these studies, deep neural networks were used to implement more complex temporal models. Early attempts frequently compared time-series methods based on classical nonparametric and parametric models with the abilities of neural architectures. Shawon et al. [27] applied DNN on load data from nine large cities of Bangladesh and achieved $R^2 = 0.991$ and MAPE = 0.011 that significantly outperformed the traditional benchmarks. Mahjoub et al. [28] compared LSTM, Multi-layer GRU and Drop-GRU using average

consumption data from Péronne, France, reporting the best performance with LSTM for 1, 3, 7 and 15-day forecasts with RMSE = 0.0508, 0.0904, 0.0844 and 0.0837. Abumohsen et al. [29] studied on Tubas Electricity Company, Palestine hourly load data (weather, weekdays, months), where GRU performed the best ($R^2 = 90.228\%$, MSE = 0.00216, MAE = 0.03266). Ampimah et al. [30], for example, compared ARIMA and ANN on the basis of their performance on the daily household load data of 709 Irish homes, recorded over 18 months. The ANN model produced significantly better results than ARIMA with MAPE and MSE of 173.66 and 174.42, respectively, compared to 689.21 and 658.23 for ARIMA, demonstrating the capability of ANNs to deal with nonlinear data. Sayed et al. [26] further extended such analysis over 26,084,029 monthly consumption readings (2019–2022) of the Dubai Electricity and Water Authority (DEWA). As for medium-term (1 month ahead) prediction, both ANN and Random Forest achieved around 97 % accuracy which was superior to what ARIMA and MLR had provided. But it was unable to generalize because of the absence of physical weather attributes.

Iqbal et al. [31] furthered this line with a hybrid BiLSTM-BiGRU architecture with a connected layer and model bidirectional dependencies. For load and generation test datasets, it achieved MSE of 0.0058 (load) and 0.0033 (generation), which confirms the superiority of hybrid RNNs. Son et al. [32], in the same year, employed LSTM and DNN for time-series based medium and long-term power load forecasting in the Naju-si city of Jeonnam province of South Korea. In their case, DNN was documented to outperform LSTM with MAE of 0.02 and 0.05, RMSE of 0.04 and 0.08, MAPE of 0.21 and 0.48 and identical R^2 of 0.99.

Recognizing the limits of standalone deep learning, researchers began integrating feature selection and optimization. Memarzadeh et al. [33] applied Wavelet Transform to Mutual Information-Interaction Gain (MI-IG) for closed-loop next day forecasting, with MAPE of 0.29 % (PJM dataset) and 0.55 % (Austrian dataset). Hafeez et al. [34] reported THEMIE-PO-GWDO (Themis Prediction Octadic-Product with GWDO) using on U.S. grids FE (First Energy dataset), EKPC (East Kentucky Power Cooperative dataset) and Dayton. This method chose the best features and accelerated training, which yielded state-of-the-art performance compared with MI-ANN, AFC-ANN, Bi-Level and LSTM. Chen et al. [35] also proposed a hybrid CEEMDAN-IGWO-GRU model (combining CEEMDAN with iterative group-wise optimization and gate recurrent unit network) for short-term load forecasting, which can effectively suppress load fluctuation and improve prediction accuracy. The experimental results show that the CIG model achieves high precision with MAPE of 0.6997 %, RMSE of 52.4685 and MAE of 38.1891 MW, outperforming other algorithms in terms of these metrics.

Liu et al. [36] used Fuzzy C-Means clustering with LSTM on the GEFCom2014 and North American Utility datasets (14 years of data) obtaining MAPEs of 2.14 (overall) and 2.65 (max load) for GEFCom2014; and 1.51 (overall) and 0.56 (max load) on the North American dataset. Noaman et al. [37] introduced LSTM-GRU hybrid for Indian consumption data (2019–2020), resulting in MAE = 0.004, RMSE = 0.01. Somu et al. [38] developed an LSTM model trained by the sine cosine optimization and named it as eDemand, which enhances predictions under short, mid and long-term scenarios. Despite significant progress in optimization methods, developing lightweight and scalable algorithms that effectively trade-off between exploration and exploitation remains a challenge [39].

Attention mechanisms were incorporated to emphasize an important content in a sequence. Lin et al. [40] proposed to use an attention mechanism over LSTM and achieved 6.5 % better than the state-of-the-art models. Wang et al. [41] generalized to BiLSTM with Rolling Updates on three Australian datasets and expressed an increase in accuracy, lowered RMSE and reduced computation time. Zang et al. [42] proposed a self-attention-based LSTM method for 27 households in Jiangsu, China by considering weather feature (temperature, humidity, wind speed) and outperformed ARIMA, SVR, ANN and LSTM.

Buratto et al. [43], who introduced Seq2Seq LSTM with attention to

Brazilian load data (2021–2023) and achieved 0.3027 in MAE. Jung et al. [44] used attention-based GRU on both U.S. and Korean building datasets, with an improvement over the conventional GRU and LSTM of 10 %. Liao et al. [19] integrated TPA with LSTM using Panama 2015–2020 national load data, including economic and climate indexes, to obtain MAE of 51.43 MW, RMSE of 73.73 MW and MAPE = 4.41 %. Lu et al. [45] proposed online correction to hybrid LSTM, considering Belgium, Denmark and Norway datasets (2019–2021) and the MAEs were 255.718 MW, 42.55 MW and 339.112 MW, remarkably enhancing the robustness.

CNNs, which are good at capturing local patterns, have been extensively, combined with RNNs. Wu et al. [46] introduced the CNN-GRU method for short-term load forecasting in Wuwei, China (31 days, 44, 640 samples) with MAPE of 2.8839 % and a RMSE of 1203.23 MW, which is more accurate than BPNN and individual models. Aslam et al. [47] proposed the Convolutional NN with CHIO (Corona virus Herd Immunity optimization) and tested it in ISO—NE data, which has achieved the load and price prediction accuracy of 95 % and 90.789 %, respectively. Yi et al. [18] combined self-attention with CNN-LSTM and applied it to Australian Government and IEEE Dataport datasets, with MAPEs of 0.351 (residential) and 0.027 (non-residential), respectively.

Li et al. [48] in 2023, considered the sizeable characteristics of Integrated Energy System (IES) in terms of volume of data, randomness and coupling of multi-source energy. The study analysed the multi-energy power demand characteristics from the data provided by Arizona State University in 2020 and was able to achieve 98.01 % accuracy with lowest MAPE of 1.97 %, both, during winter.

The trend, lately, favours interpretability with XAI, for its ability to provide deeper and meaningful insights into the working of a model. In 2023, Nooruldeen et al. [49] studied on employing single and ensemble ML algorithms in the prediction of daily loads for Kirkuk, Iraq. The algorithms included AdaBoost, Bagging, SVR and DT. Considering feature deduction, PCA and LDA had been implemented. Through all analysis it was confirmed that bagging algorithms performed superior outperforming AdaBoost, SVR and DT having reduced squared difference and deviation. Sarker et al. [50], also advocated for integration of XAI in AI based smart grid load prediction tasks. The proposed attention-based 1D-CNN-GRU model highlighted the importance of feature analysis, using SHAP analysis and hyperparameter tuning, using PSO and achieved MAE of 0.12, 0.8, 16.48 and 82.64 across 4 different datasets. Maarif et al. [51], used a public energy usage dataset from “Korea Electric Power Corporation” and proposed a model using single-layer LSTM, double-layer LSTM and bi-directional LSTM supported by XAI. The authors claim the analysis made by XAI to bring transparency to the proposed model and boasts high interpretability of the architecture with accurate results in terms of RMSE scores by up to 0.08, 0.07 and 0.07 for the three versions of LSTM, respectively. They predicted the Hourly energy usage with such good performances.

The hybrid LSTM-CNN architecture has been increasingly adopted in energy forecasting tasks due to its ability to model both temporal dependencies and local-spatial variations in load data. LSTM networks are effective for learning long-term time-series patterns, while CNNs excel at extracting spatial features and identifying short-term fluctuations [52]. Ibrahim et al. [53] demonstrated improved forecasting of solar power generation using CNN-LSTM, while Sharma et al. [54] showed its effectiveness in household load prediction. Salman et al. [55] further validated that the hybrid approach outperforms standalone models in capturing complex consumption behaviours. XAI further enhances the interpretability of hybrid LSTM-CNN models by providing insights into the internal decision-making processes of these otherwise opaque architectures [56]. Techniques such as IGA and SHAP enable the identification of influential input features, allowing researchers and energy managers to understand which features drive specific forecasts. But very few studies have integrated XAI with segmented (day peak, evening peak and off-peak) data, which provide deeper meaningful analysis and interpretability of the models. This leaves a significant gap in generating

tangible results that can be generalized to larger and diverse grids. This study aims to develop an electric load forecasting architecture capable of supporting large-scale applications. Given the segmented nature of energy demand in this study, the CNN-LSTM with IGA-XAI model provides a robust and accurate framework for forecasting urban load dynamics in smart grid application.

The Table 1 shows an overview of the significant works done in the recent past on the domain of load forecasting across various locations in different time periods. The studies conducted within the recent years, clearly lack in incorporating peak and off-peak periods' load data through the day. This study proposes to address all these above-mentioned challenges by considering day peak, evening peak and off-peak load data of the city of Chattogram.

While numerous studies have employed hybrid CNN-LSTM architectures for load forecasting, including enhancements such as attention mechanisms [18], few have addressed the temporal segmentation of energy demand into distinct peak and off-peak periods within urban settings. Furthermore, existing approaches often treat the forecasting process as a "Black Box", lacking transparency in the models' decision-making. In contrast, this study integrates XAI technique, like IGA, to provide detailed attributions of input features across the segmented dataset, which allows for a deeper understanding of the proposed model's behavior during Day Peak, Evening Peak and Off-Peak periods. By revealing how specific features influence predictions at different times, XAI enhances both the interpretability and transparency of the model, making it more reliable for practical deployment in energy management systems.

3. Methodology

This section outlines the approaches used to evaluate the effectiveness of machine learning models in energy consumption forecasting as

well as description of the data and steps taken during the preprocessing of the obtained data. The reasons behind selecting these models were their wide application in regression analyses. They also differ in their approach to learn and predict data. The proposed process allows a step-by-step evaluation of the quality of the model's forecast when applied to different regions. With this critical approach, we propose to establish a framework which can be utilized for better forecasting of energy demand, leading to informed decisions on energy management and planning. The workflow of the proposed model has been illustrated in Fig. 1.

The machine learning pipeline presented in this work is set up to maintain a systematic and reproducible approach during each step of the modeling pipeline, going all the way from data fetching to model evaluation and workflow. Fig. 1 gives an overview of the proposed approach, and Fig. 2 shows a refined flow of tasks that has been followed during model development and evaluation. This flow chart becomes a frame of the logical diagram, in terms of data collection, exploratory data analysis, data cleansing, feature engineering, model fitting and performance assessment. It allows to clarify the relation between the single steps, and how the choices at one step impact the following steps in the workflow.

With this abstracted workflow in mind, the next section describes further the data pre-processing steps taken in this project. These stages define the basis for the proposed methodology, and are critical to be able to train the models with trusted, representative and properly structured input data.

3.1. Data pre-processing

Data pre-processing is one of the most important sections of machine learning. It deals with clearing abnormal data and transforming the dataset into an optimized and high performing one for the algorithms to accurately find relations. This section elaborates on the description of data, normalization and percentage of train-split data.

Table 1
Brief overview of recent studies on energy forecasting.

Reference	Year	Dataset Description	Models used	Performance Analysis	Limitation(s)
[46]	2020	44,640 samples across 31 days from Wuwei, Gansu Province of China	GRU-CNN	MAPE: 2.8839 % RMSE: 1203.23	Limited factors, data, training efficiency
[32]	2020	Hourly data from two industrial buildings, from 2017 to 2019, in the Naju-si city of Jeonnam province of South Korea	DNN	MAE: 0.02 RMSE: 0.04 MAPE: 0.21 R ² : 0.99	No weather data, limited model comparison
[26]	2021	26,084,029 records for monthly electricity consumption of Dubai, from 2019–2022	ANN, RF	Accuracy: 97.1 %	Lack of real weather data integration
[49]	2023	Daily loads for Kirkuk, Iraq	Bagging Model using PCA with 8 Features	MSE: 63.54 % RMSE: 7.97 % MAPE: 8.48 %	Dataset-specific results, limited real-time factors
[30]	2023	Daily electricity load data from 709 individual households in Ireland for a period of over 18 months	ANN, ARIMA	ANN: MAPE: 173.66 % MSE of 174.42 ARIMA: MAPE: 689.21 % MSE: 658.23	Missed peak loads, limited model comparison
[19]	2023	All region historical load data in Panama	TPA-LSTM	MAE: 51.43 MW RMSE: 73.73 MW MAPE: 4.41 %	Short term only, Limited sampling integration
[18]	2023	Australian Government Department of the Environment and Energy and IEEE Dataport.	LSTM-CNN with self-attention	MAPE: 0.351 (Residential) MAPE: 0.027 (Generalized)	Randomness, weak nonstationary handling.
[48]	2023	Arizona State University in 2020	CNN	Accuracy: 98.01 % MAPE: 1.97 %	Missing external data, low-resolution data.
[51]	2023	Public data from Korea Electric Power Corporation	single-layer LSTM, double-layer LSTM, bi-directional LSTM	RMSE: 0.08, 0.07, 0.07	High computation cost, low interpretability trade-off
[43]	2024	Brazilian dataset with load data from March 2021 to April 2023	Seq2Seq-LSTM-Attention	MAE: 0.3027	Needs hyperparameter tuning, high complexity
[50]	2024	4 datasets: Panama Case Study Dataset, Daily Electricity Price and Demand Dataset, Household Electric Power Consumption Dataset, Australia Load Dataset	Attention-based 1D-CNN-GRU	MAE: 0.12, 0.8, 16.48, 82.64	Limited feature exploration and preprocessing

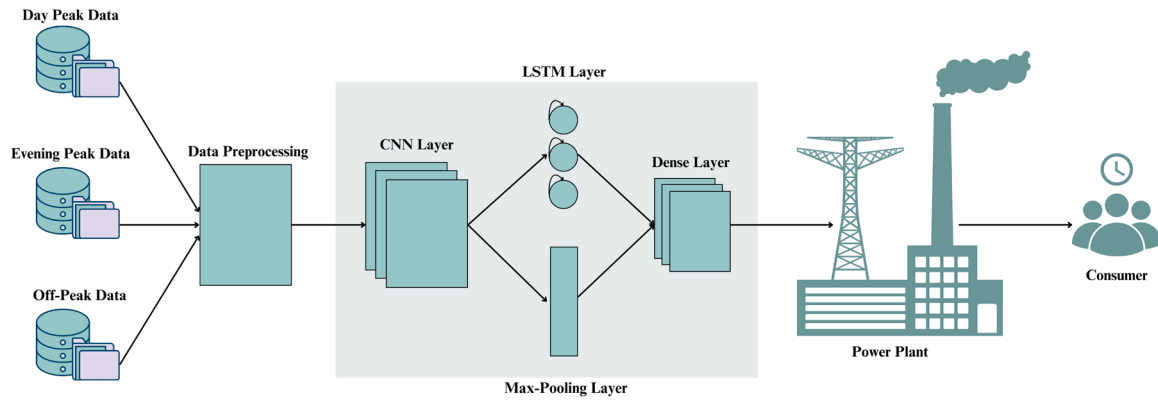


Fig. 1. Proposed workflow diagram of the study.

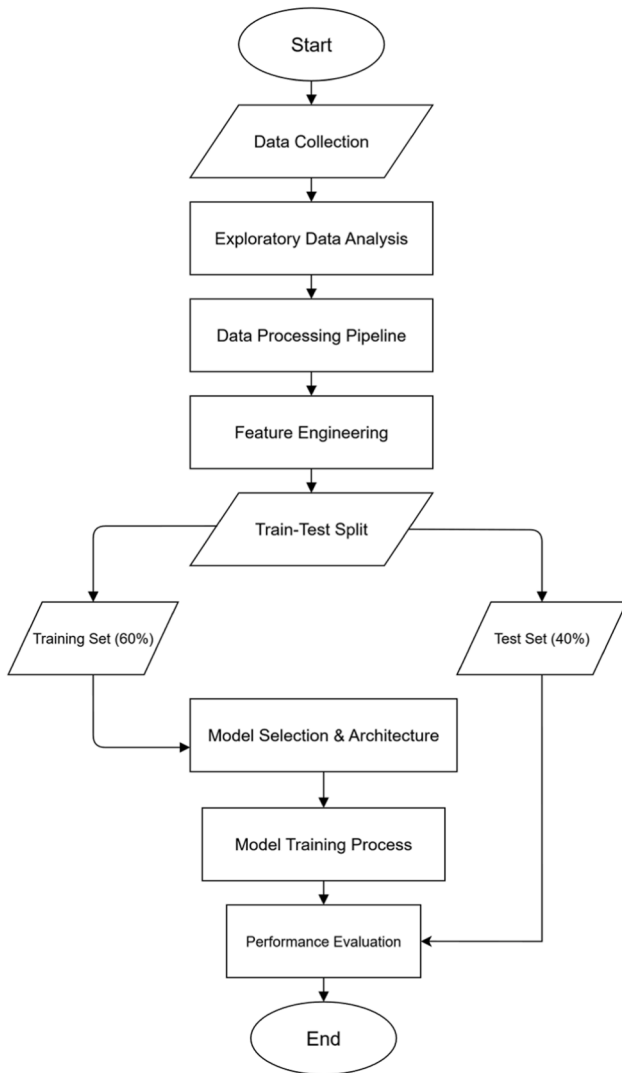


Fig. 2. Flowchart of the proposed ML workflow.

3.1.1. Dataset description and correlation analysis

The dataset had been collected from Chattogram office of the Power Grid Company of Bangladesh (PGCB). It contains the load demand of Chattogram starting from 1st of January 2021 until 30th June 2023. It consists of 3 temporal segments namely 'Day Peak (9am-5pm)', 'Evening Peak (6pm-9pm)' and 'Off-Peak (10pm-8am)' which contains the peak

load demand (MWh) in Chattogram during the day, evening and the off-peak respectively. The 'Temperature', 'Dew/Frost Point', 'Wet Bulb Temperature', 'Earth Skin Temperature', 'Min Temperature', 'Max Temperature', 'Specific Humidity', 'Relative Humidity' and 'Precipitation' of Chattogram that had been collected from the NASA Prediction of Worldwide Energy Resources (<https://power.larc.nasa.gov/>) website and added to the dataset. Temperature, throughout studies have acted as one of the main determiners of electric load demand, especially in temperate regions [57]. Earth skin temperature also shows positive relation with electricity consumption in the temperate regions [58]. Other weather factors like dew point and relative humidity also affects electricity consumption [59]. Specific humidity has also proven to be crucial in this particular task [60]. Precipitation has also proven to be a crucial factor in forecasting energy demands [61]. Wet bulb temperature, similarly shows a positive relation with energy consumption in warmer climate [62].

A correlation analysis was conducted to investigate the linear relationships between various meteorological parameters and the daytime peak energy demand (Day Peak), as illustrated in Fig. 3. The Pearson correlation matrix reveals that temperature-related variables exhibit the strongest positive correlations with peak load. Specifically, ambient temperature ($r = 0.75$), wet bulb temperature ($r = 0.76$), earth skin temperature ($r = 0.74$) and dew/frost point ($r = 0.73$) all show strong linear associations with daytime peak demand. These results indicate that elevated thermal conditions significantly drive energy consumption, likely due to increased cooling requirements in urban settings. Furthermore, specific humidity ($r = 0.72$) and minimum temperature ($r = 0.76$) also demonstrate notable positive correlations, suggesting that thermal discomfort induced by humidity may contribute to additional energy usage during peak hours. In contrast, relative humidity ($r = 0.27$) and precipitation ($r = 0.19$) show weak correlations with the Day Peak load, implying a more limited or potentially nonlinear influence on electricity consumption.

The relatively weak linear relationships of variables such as precipitation and relative humidity with peak load further justify the use of deep learning models, particularly convolutional neural networks (CNNs), which are capable of capturing complex nonlinear and temporal interactions that traditional correlation analysis may overlook. These findings were foundational in the design of the proposed hybrid CNN-LSTM model, enabling it to focus on the most influential and non-redundant meteorological features for accurate mid-horizon peak load forecasting.

Following the day peak correlation analysis, a similar trend is observed in the evening peak load correlation matrix illustrated in Fig. 4. Temperature-related features like ambient temperature ($r = 0.82$), wet bulb temperature ($r = 0.83$) and earth skin temperature ($r = 0.81$) continue to exhibit strong positive correlations with the evening peak load, indicating that thermal conditions remain a dominant driver of energy demand throughout the day. Specific humidity ($r = 0.76$) and

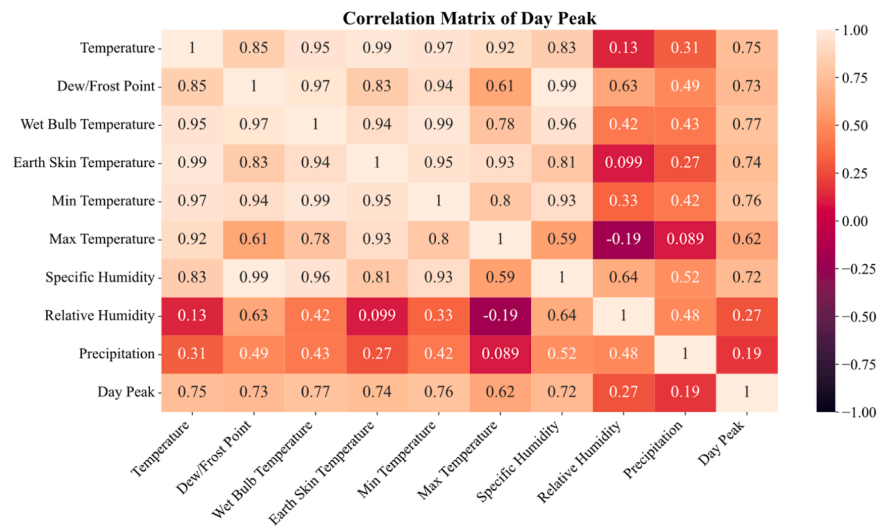


Fig. 3. Correlation matrix for day peak data.

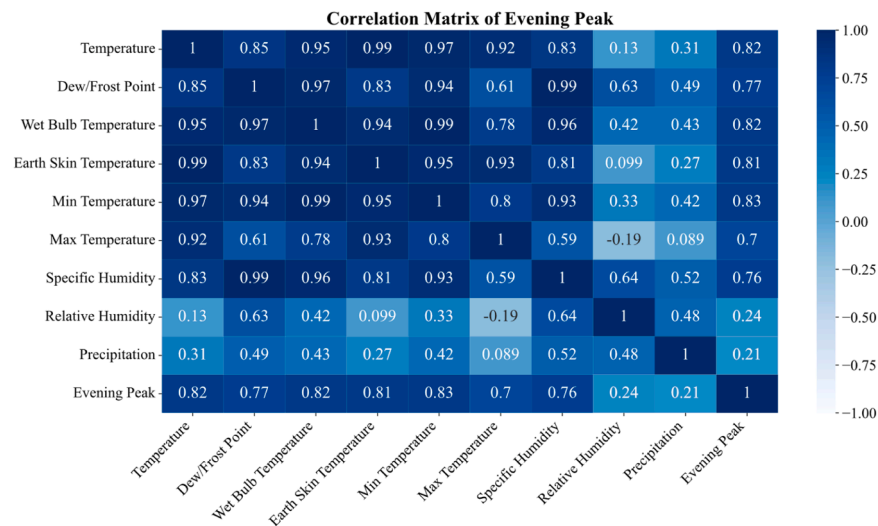


Fig. 4. Correlation matrix for evening peak data.

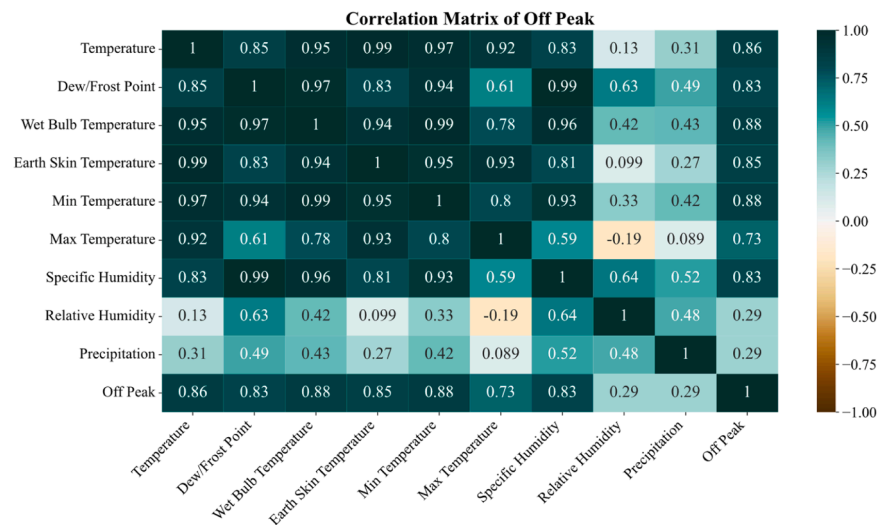


Fig. 5. Correlation matrix for off-peak data.

minimum temperature ($r = 0.79$) also shows notable associations, further reinforcing the role of retained heat and moisture in shaping evening consumption patterns. As with the day peak, relative humidity and precipitation presents weak correlations ($r = 0.21$), suggesting limited direct linear influence. Overall, the evening peak correlation structure reflects similar patterns to that of the day peak, albeit with slightly stronger coefficients, pointing to a higher meteorological dependency during evening hours which is likely due to delayed thermal dissipation in urban environments and sustained cooling needs in residential areas.

Extending the correlation analysis to off-peak periods, we observe a consistent pattern with the day and evening peak matrices, particularly in the influence of temperature-related variables. As shown in Fig. 5, temperature-related variables such as ambient temperature ($r = 0.86$), wet bulb temperature ($r = 0.88$) and earth skin temperature ($r = 0.83$) continue to show strong positive correlations with energy load, reaffirming their dominant role across all time intervals. Specific humidity ($r = 0.83$) and minimum temperature ($r = 0.80$) also maintain high associations, similar to those observed during evening peaks. While relative humidity and precipitation remain weakly correlated ($r = 0.29$), their slightly higher relevance during off-peak hours suggests a modest baseline influence. Overall, the off-peak correlation structure closely mirrors the peak periods, further supporting the use of weather-sensitive features in the proposed CNN-LSTM model for robust, time-generalizable load forecasting.

The correlation analyses across day peak, evening peak and off-peak periods consistently highlight the strong influence of temperature and humidity-related variables on urban energy demand. These patterns justify the need for a model capable of capturing both spatial feature dependencies and temporal dynamics. The proposed 1D-CNN-LSTM hybrid model effectively addresses this by leveraging CNN to extract key patterns from meteorological inputs and LSTM to model time-based dependencies. This combination enables accurate and robust load forecasting across different periods, making it well-suited for complex, climate-sensitive urban environments.

3.1.2. Data normalization

Standard scalar has been used to standardize and generalize the processed dataset. For machine learning algorithms, particularly those which are sensitive to fluctuations in feature scales, standard scaler is used to standardize the input data in a way that ensures that the data points have a proportional and consistent scale. The altered features usually have a mean of 0 and a standard deviation of 1 after standardization. Hence, rather than containing a constant range, the standardized feature set is centered about 0 with a variance of 1.

3.1.3. Train-test split

A 70–30 split was initiated where 70 percent of the total dataset had been taken for training the model. The rest 30 percent were kept and used for validation, ensuring for the prevention of both overfitting and underfitting of the obtained model.

3.2. Machine learning models

There were suggestions that the use of CNN was for its ability to learn complex patterns and hence it was chosen as the initial method. However, considering the time series nature of the dataset, RNN models were then used. Since standard RNNs have vanishing and exploding gradients as known issues, GRU algorithm was preferred because of its computational efficiency and performance in time series tasks.

Afterwards, LSTM networks were highlighted as they are designed to capture long-term dependencies and have had good results in other related researches. It was also stated that transformers combined with LSTM had outperformed others under specific circumstances thus a hybrid model based on this combination was set up for now.

Quite often, there may be difficulty in forecasting peaks accurately

by using LSTM models which led to the application of an alternative advanced neural network method; CNN. CNNs were thought to be especially good at capturing spatial hierarchies in data, which could enhance forecasting accuracies. These algorithms have been individually discussed in details below.

3.2.1. Convolutional Neural Network (CNN)

CNNs utilize convolutional layers, which analyzes the spatial hierarchies of features from input data. As a result, convolutions that occur in these layers are involved in capturing local patterns while reducing the number of parameters than fully connected networks do. Additionally, pooling layers help to reduce the dimensions of spatial data and control overfitting among others.

Convolutional layer, with which the model begins, is given below; where the output feature map of the l -th convolutional layer was denoted as F_l and $F_{l,i,j,k}$ denotes the position of activation of the k -th layer.

$$F_{l,i,j,k} = \sigma \left(\sum_{m=1}^{M_l} \sum_{n=1}^{N_l} x_{l-1, i+m-1, j+n-1, c} \cdot w_{l,m,n,c,k} + b_{l,k} \right) \quad (1)$$

Here, M_l and N_l denote the l -th filter's dimensions, x_{l-1} denotes pervious layer's inputs, w_l denotes the weight of the filters, b_l denotes the bias term and the ReLU activation term is denoted by σ .

3.2.2. Long Short-Term Memory (LSTM)

Long Short-Term Memory addresses the vanishing gradient problem, of traditional RNNs, by incorporating memory cells that can retain information over long periods. LSTMs have three types of gates: input, output and forget gate. This is in respect to long-term dependencies that are often encouraged in this network, in order to capture the temporal dependencies in sequential data.

The LSTM model has been mathematically explained below:

Forget layer:

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f) \quad (2)$$

Here, σ is the sigmoid activation function, W_f is the weight matrix associated with the forget gate, h_{t-1} is the hidden layer from the previous time step, x_t is the current time step input and b_f is the bias term.

Input layer:

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i) \quad (3)$$

Here, W_i is the weight matrix associated with the input gate and b_i is the bias term.

Hidden layer:

$$h_t = o_t * \tanh(C_t) \quad (4)$$

Here, h_t is the hidden state, C_t is the updated cell state, $\tanh$ is the hyperbolic tangent activation function and o_t is output gate.

3.2.3. Proposed hybrid 1D-CNN-LSTM model

The proposed model was an attempt to unify the spatial and temporal intricacies within the dataset. This led to the selection of 1D-CNN for the detection of local spatial patterns and capturing of short-term dependencies within the data, merged with LSTM, that is capable of retaining information over long sequences, addressing the temporal nature of the data. The first part of the architecture contains the 1D-CNN structure, which seeks to understand the input sequences by self-learning numerous small, yet essential, features over repetitive interacting regions utilizing a number of convolutional layers.

Such layers, illustrated in Fig. 6, are followed with a Global-MaxPooling1D layer, which helps collapse the data into a fixed-dimensional vector by keeping only the most notable details. In parallel with this, an LSTM layer takes care of the sequential data coming

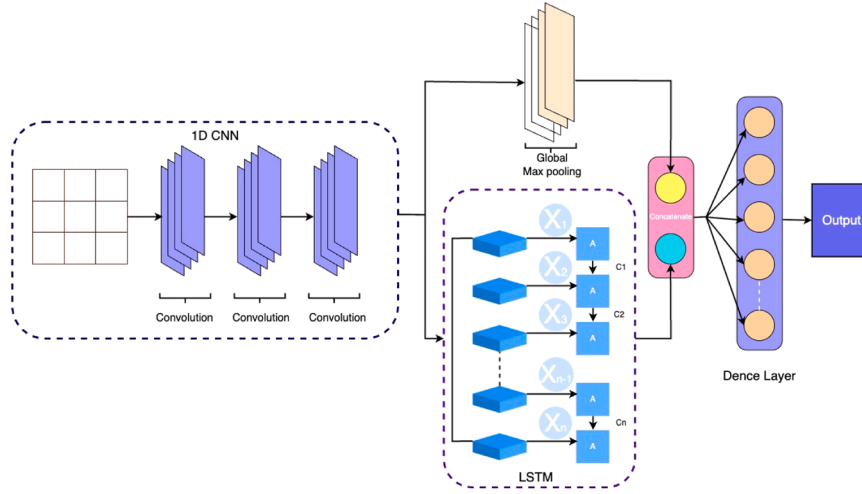


Fig. 6. Architecture of the proposed hybrid 1D-CNN-LSTM model.

from the last CNN layer to encode long-range information. Then, the outputs from both the CNN and the LSTM layer, are passed through the fully connected layers after merging the two outputs and the final forecasting is done. The addition of these layers allows the model to consider both the local structure as well as temporal structure, which enhances the ability of the model to perform forecasting. By combining these approaches, the model will be able to handle challenging data with time series pattern better than using either CNN or LSTM solutions separately, thereby increasing forecasting accuracy.

The hybrid 1D-CNN-LSTM model was then tested on combinations of the computational parameters. This allowed for better adaptability of the hybridized model into the data provided. Upon rigorously testing all the hyperparameter configuration combinations, the final tuning was done to provide stability to the proposed model. The model showed little variance in different configurations and the best results obtaining hyperparameter configuration has been shown in Table 2.

In the proposed model, 3 convoluted layers were utilized with filter sized 32, 64 and 64, respectively, with the kernel size of 3 and ReLU selected as the Activation function. These layers focus on local feature extraction of meaningful spatial patterns. Then, a GlobalMaxPooling layer was introduced to identify and enhance the most impactful features. The LSTM layer used afterwards, captures the temporal dependencies by forgetting the insignificant readings with a dropout of 0.1 in the recurrent layers, with the dense layers set to 32 units. This overall tuning of the hybrid model creates a reliable architecture capable of forecasting the intricacies in the data and following the load pattern across all the peaks and off-peaks.

Table 2

Tuned hyperparameters of improved CNN-LSTM framework.

Component	Hyperparameter	Value
Input Layer	Input Shape	(WINDOW_SIZE, 10)
Conv1D Layers (1–3)	Filters	32 + 64 + 64
	Kernel Size	3
	Activation	ReLU
	Padding	Same
	Type	GlobalMaxPooling1D
LSTM Layer	Units	32
	Recurrent Dropout	0.1
Concatenation	Layers	cnn_output, lstm_output
	Units	32
Dense Layer 1	Activation	ReLU
	Units	32
Output Layer	Units	1

3.3. Evaluation metrics

In the validation of the proposed algorithms and models, various evaluation matrices had been performed. These matrices have been discussed in detail in this section.

3.3.1. Mean Squared Error (MSE)

The Mean Squared Error (MSE), also known as Euclidean squared errors, is a performance metric used to assess the quality of estimators in square measurements. This measure always remains positive and reduces with decreasing estimation errors. This method is indeed a valuable tool for assessing model performance especially in regression analysis.

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 \quad (5)$$

Here,

n = number of data points
 Y_i = observed values
 $\hat{Y}_i$ = Forecasted values

3.3.2. Root-Mean-Square Error (RMSE)

RMSE calculates average discrepancy between forecasted values generated by statistical models and actual observed values. RMSE scales down mean squared errors into square roots thereby giving accuracy metrics within target variable units that facilitate interpretations on models' performance.

$$RMSE = \sqrt{\frac{\sum_{i=0}^N (x_i - \hat{x}_i)^2}{N}} \quad (6)$$

Here,

i = variable i
 N = number of non- missing data points
 x_i = actual observations time series
 $\hat{x}_i$ = estimated time series

3.3.3. Mean Absolute Error (MAE)

This is an arithmetic mean which computes the settled deviations between two co-related data points such as forecasted values and real occurrences or outcomes of alternative measuring techniques. It sums these differences up and thus provides a brief summary of how far off the

forecasting was.

$$MAE = \frac{\sum_{i=1}^n |y_i - x_i|}{n} \quad (7)$$

Here,

y_i = forecasting
 x_i = true value
 n = total number of data points

3.3.4. Mean Absolute Percentage Error (MAPE)

A forecasting method's accuracy is evaluated with MAPE, a statistical measure. It computes the mean absolute percentage error between the forecasted values and the actual ones. Since it shows how far wrong forecasting is from real data, MAPE serves as an easy-to-understand quantitative metric of forecast quality that can be expressed in terms of percent. Because this measure is normalized to a scale of data, it allows for comparing accuracy among different datasets or models and thus becomes an indispensable instrument for performance estimation in various predictive modeling settings.

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{A_t - F_t}{A_t} \right| \quad (8)$$

Here,

n = number of times the summation iteration happens
 A_t = actual value
 F_t = forecast value

3.3.5. Coefficient of determination (R^2)

The coefficient of determination or R^2 is a statistic for regression analysis which seeks to show how much variability in the dependent variable is accounted for by independent variables. It shows the percentage of variance in the output that can be explained with respect to predictors and serves as an index of model's explanatory capability.

$$R^2 = 1 - \frac{RSS}{TSS} \quad (9)$$

Here,

RSS = sum of squares of residuals
 TSS = total sum of squares

3.4. Noise injection for certainty analysis

The proposed model uses a significant set of weather-related features in a multivariate time series to understand the association of the meteorology with the variable of interest. Weather reports and forecasts, while having valuable attributes, face uncertainty because of the chaotic nature of the atmosphere, the stochastic nature of instruments, and possible errors in measurements taken, even in collection and forecasting. Therefore, prediction models relying heavily on such information, often struggle in real-life operational environments where input data is heavily corrupted.

Inspired by the work by Zhang et al. [63] the addition of controlled noise to test data works as a practical stand-in for measuring a model's robustness and generalization capability. Performance variability for metrics during noisy conditions allows us to measure the model's sensitivity to perturbations in its input. This variability is, in essence, an application of a model's robustness to disturbances and its ability to behave in a realistic, uncertain world in a controlled, predictable manner.

To study the performance of our model in such conditions, we have designed a controlled scenario of noise injection. In particular, Gaussian

distributed noise was in the weather features inputted at the levels of 5 %, 10 %, 15 % and 20 %. The aim here is to demonstrate realistic sensor errors, data not collection, and a range of lackluster oscillation in forecasts through potential distortions. Incrementally increasing the noise level allows us to observe the degradation tendencies the model exhibits, thereby evaluating the model's sensitivity to the corrupted data it receives and the toleration threshold it has before the predictive outcome suffers marked degradation.

$$p(z) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(z-\mu)^2}{2\sigma^2}} \quad (10)$$

z = the random variable (feature value after noise injection).
 μ = the mean (expected value) of the distribution.
 σ = the standard deviation, which controls the spread (how wide or narrow the distribution is).
 σ^2 = the variance (square of the standard deviation), which measures the dispersion of the distribution.
 $p(z)$ = the probability density of observing the value under this Gaussian distribution.

The sensitivity analysis, or robustness analysis, serves a dual function. It helps quantify the level of uncertainty in the model, which in turn helps assess how confidently the model can predict an outcome when it is wrong given uncertain, data-challenged conditions. It also describes the reliability and stability of the model, which is paramount for practical application especially in high-risk predictive tasks, as it describes the model's operational error being capable of causing severe resultant ramifications. With the model demonstrating noise stress testing, it helps confirm inferences drawn regarding the model being robust, as such a feature is fundamental to the ease of its adoption in practice.

4. Results and analysis

In this research, upon rigorous analysis of the relevant literature and class-leading studies, a hybrid CNN-LSTM model was implemented due to the models showing a fairly higher performance in comparison to the other forecasting models. Lag values play a vital role in predictive modeling, particularly in time series forecasting, as they help capture the sequential relationships and recurring patterns within the dataset. By incorporating previous time steps, models can better understand how historical data points impact future values. Hence, the first phase of the study consisted of finding the accurate lag values to determine and work on the best possible time-based input values. Upon testing the hybrid model on 9 corresponding lag values, the lag values showed to have impacted the results positively, which has been shown in Fig. 7.

Fig. 7 clarifies the importance of lag features in forecasting tasks. As the lag values increased, the R^2 values increased, indicating increase in the capability to follow the load trend along with the discrepancies. The negative correlation with the RMSE suggests the reduction in error, hence more accurate forecasting by the proposed hybrid model. This performance improvement can be attributed to the ability of the CNN-LSTM architecture to leverage broader temporal context as more historical datapoints (Lags) are provided. Further analysis of lag values has been given in Table 3.

The results in Table 3 instate the positive correlation of Lag values and the forecasting accuracy across Lag 1 through Lag 7. The model reaches saturation at Lag 7 and a slight decline is noticed at Lag 9. From these results, Lag 7 has been selected as the optimal parameter for further analysis, like- loss curves for the three peaks. The training loss curves in Fig. 8, shows that, for all three peaks, a saturation point is reached at around Epoch 200, when compared in terms of RMSE. It also demonstrates how the tuned hyperparameters contribute to the creation of a model that accurately captures the load pattern of a metropolitan

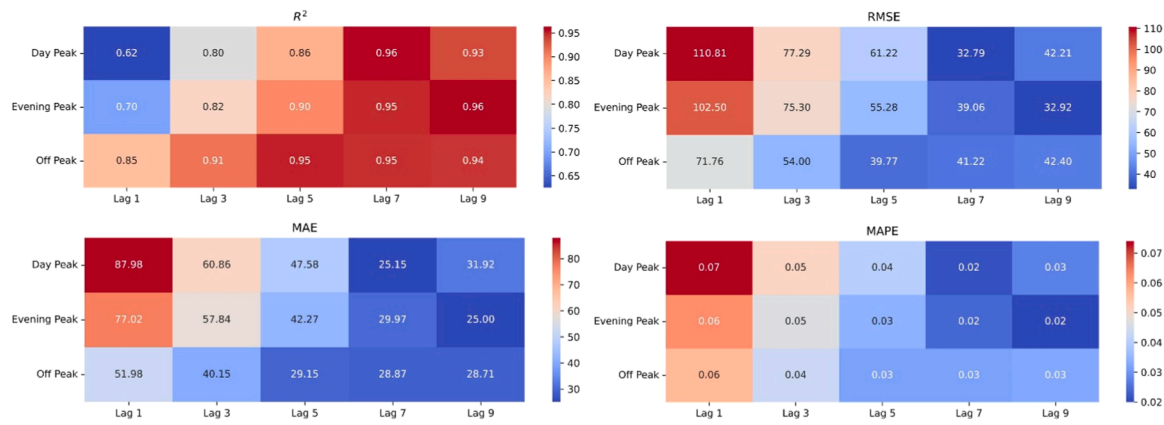


Fig. 7. Heatmap analysis for lag optimization.

Table 3
Results of CNN-LSTM for lag 1 through lag 9.

Lag	Peak	MSE	RMSE	MAE	MAPE	R ²	AVG R ²
1	Day Peak	12,278.9839	110.8106	87.9848	0.0741	0.6246	0.7234
	Evening Peak	10,505.896	102.4983	77.0196	0.0632	0.696	
	Off-Peak	5149.7301	71.7616	51.9806	0.0571	0.8496	
3	Day Peak	5974.0815	77.2922	60.8597	0.0514	0.795	0.8415
	Evening Peak	5669.3721	75.2952	57.8406	0.0465	0.8187	
	Off-Peak	2916.459	54.0042	40.1516	0.0431	0.9108	
5	Day Peak	3748.3939	61.2241	47.5778	0.0403	0.8647	0.9044
	Evening Peak	3055.3641	55.2753	42.274	0.0342	0.8979	
	Off-Peak	1581.7774	39.7716	29.149	0.0315	0.9507	
7	Day Peak	1075.0743	32.7883	25.1459	0.0213	0.9602	0.9515
	Evening Peak	1525.5945	39.0589	29.9743	0.024	0.948	
	Off-Peak	1699.2511	41.222	28.8746	0.0315	0.9464	
9	Day Peak	1781.5315	42.2082	31.9165	0.0271	0.9342	0.9469
	Evening Peak	1083.8302	32.9216	25.0035	0.0199	0.9632	
	Off-Peak	1797.4062	42.3958	28.712	0.0324	0.9434	

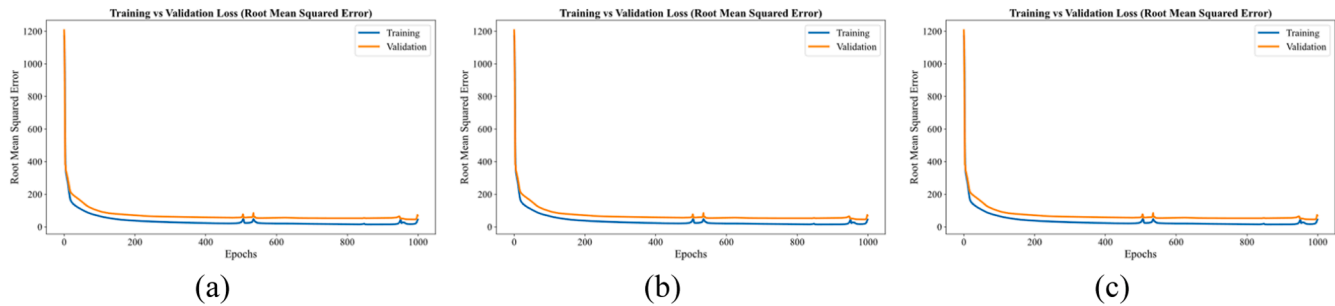


Fig. 8. Loss curve for CNN-LSTM during (a) day peak; (b) evening peak and (c) off-peak.

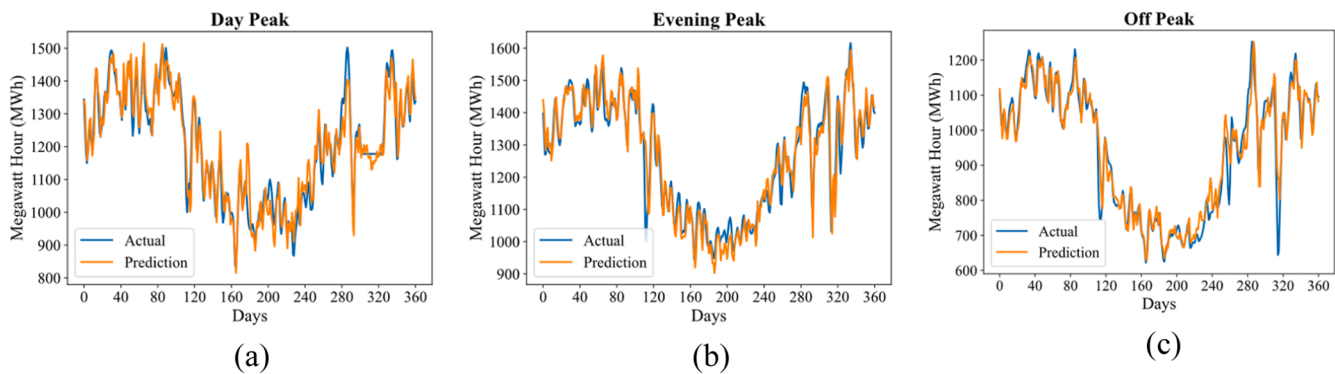


Fig. 9. Actual vs predicted for (a) day peak; (b) evening peak and (c) off-peak.

city.

Following the successive accomplishments in Lag and Loss analysis, the predicted load was compared with the actual data. Fig. 9 and Fig. 10 presents the actual versus predicted load curves and corresponding scatter plots for Day Peak, Evening Peak and Off-Peak periods, respectively. The proposed CNN-LSTM model demonstrates strong alignment with true values across all three periods, accurately capturing both high-load fluctuations and low-demand trends. The Day Peak and Evening Peak loads reach up to 1500 MWh and 1600 MWh, respectively, while Off-Peak peaks near 1220 MWh for all of which are effectively forecasted with minimal deviation. The accompanying scatter plots further validate this performance, with most data-points tightly clustered around the 1:1 line, indicating very high correlation and extremely low error variance. These results confirm the model's robustness, generalization capability and applicability for real-time load forecasting in complex urban smart grid operations.

The results obtained by the proposed hybrid model has been further evaluated against the most popular state-of-the-art models that are currently in use for load forecasting tasks.

For analysis of the performances of the hybrid CNN-LSTM model, the results of the proposed model were compared with that of the benchmark models, in terms of- MSE, RMSE, MAE, MAPE and R^2 for each peak and off-peak durations. All these comparative results have been compiled in Table 4.

Performance is assessed using standard regression metrics: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Mean Absolute Percentage Error (MAPE) and the coefficient of determination (R^2). The average R^2 across all three peak periods is also included as an indicator of model robustness and generalization.

Among all models, the proposed hybrid CNN-LSTM consistently outperforms others across all time intervals. It achieves the lowest RMSE (32.79 for Day Peak, 39.06 for Evening Peak and 41.22 for Off-Peak) and highest R^2 values (0.9602, 0.948 and 0.9464 respectively), indicating superior predictive accuracy and generalizability. Notably, the CNN-LSTM model also records the lowest MAPE across all intervals, demonstrating its reliability even under varying load conditions. Its average R^2 of 0.9515 reflects strong overall performance across diverse temporal contexts.

The standalone CNN model performs relatively well, particularly for Day Peak and Evening Peak periods, but falls short in off-peak accuracy due to limited capacity to model long-term dependencies. LSTM and GRU show moderate performance, with LSTM slightly outperforming GRU during Day Peak but significantly underperforming during Evening Peak and Off-Peak. The GRU model, in particular, struggles with off-peak load forecasting, resulting in the highest RMSE (149.48) and lowest R^2 (0.8759), indicating its limitation in capturing more volatile or less structured demand patterns.

The Transformer-LSTM hybrid performs marginally better than GRU for off-peak predictions but underperforms relative to CNN-LSTM,

especially during Day Peak and Evening Peak intervals. Its average R^2 of 0.9098 suggests less consistent generalization across periods.

Overall, the empirical results validate the superiority of the 1D-CNN-LSTM architecture, particularly in leveraging short-term temporal dependencies (via lag-7 input) and spatial correlations among input features. This demonstrates its suitability for accurate and robust urban energy load forecasting across diverse daily demand cycles.

Proceeding forward, the day ahead forecasting was tested to find out the optimum operational range of the proposed hybrid CNN-LSTM model, which has been shown in Table 5.

Table 5 summarizes the performance of the proposed CNN-LSTM model for multi-day ahead energy load forecasting, across three daily intervals: Day Peak, Evening Peak and Off-Peak, evaluated over a 7-day forecast horizon. Standard evaluation metrics including MSE, RMSE, MAE, MAPE and R^2 are reported for each forecast day, with an additional Average R^2 (AVG R^2) indicating overall forecasting quality for each day-ahead scenario.

On Day 1, the model performs exceptionally well across all intervals, with low RMSE values (32.79 for Day Peak, 39.06 for Evening Peak and 41.22 for Off-Peak) and high R^2 scores (≥ 0.946), reflecting excellent short-term predictive accuracy. This confirms the CNN-LSTM model's effectiveness in capturing short-range temporal patterns.

As the forecasting horizon extends, a gradual decline in performance is observed. By Day 3, while the model maintains reasonable accuracy (Day Peak: RMSE = 43.76, R^2 = 0.8252), the Evening and Off-Peak predictions begin to deteriorate more noticeably, particularly in the Off-Peak period, where R^2 drops to 0.7357. From Day 4 onwards, the degradation becomes more pronounced, especially in Day Peak predictions, which show a steep drop in R^2 (0.7292 by Day 5 and as low as 0.5957 by Day 7), along with significant increases in RMSE and MAPE.

By Day 6 and Day 7, performance across all interval declines, with RMSE values exceeding 120 for Day Peak and 100+ for Evening and Off-Peak and R^2 values falling below 0.6, particularly in Day Peak forecasts. Despite this, Off-Peak predictions remain relatively more stable compared to Day Peak, suggesting that the model is better at capturing baseline demand patterns than high variability peak loads over longer horizons.

Overall, these results highlight that the CNN-LSTM model excels in short- to medium-term forecasting (1–3 days ahead) with strong generalization, while predictive accuracy understandably diminishes in longer-range forecasts (5–7 days ahead) due to the compounding uncertainty in both temporal patterns and exogenous meteorological inputs. Nevertheless, the model maintains practical forecasting utility up to Day 4, making it a robust solution for operational planning in metropolitan energy systems.

This study dove deeper proving the generalizability of the hybrid CNN-LSTM model, by testing the model's performance across two other major metropolitan cities of Bangladesh - Dhaka and Sylhet. On the data collected from January through April of 2025, from the PGCB website,

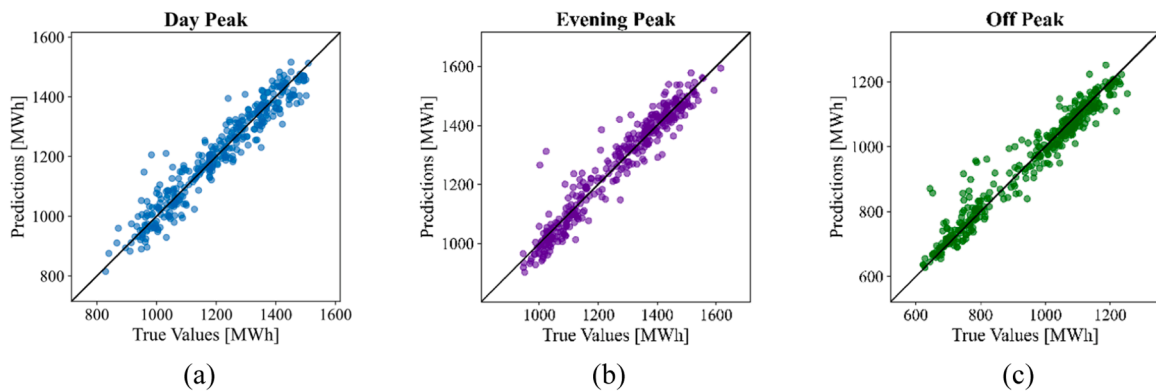


Fig. 10. Scatter plot for (a) day peak; (b) evening peak and (c) off-peak.

Table 4

Comparison results of the forecasting models using lag value of 7.

Model	Peak	MSE	RMSE	MAE	MAPE	R ²	AVG R ²
CNN	Day Peak	1739.48	41.71	36.96	0.0277	0.9514	0.9473
	Evening Peak	1585.46	39.82	29.57	0.0218	0.947	
	Off-Peak	2497.14	49.97	33.84	0.0344	0.9435	
GRU	Day Peak	4261.32	65.28	44.45	0.032	0.9213	0.9064
	Evening Peak	5505.18	74.2	51.36	0.0364	0.9221	
	Off-Peak	22,346.09	149.48	108.29	0.1117	0.8759	
LSTM	Day Peak	1770.42	42.07	37.98	0.0298	0.9514	0.934
	Evening Peak	6295.42	79.34	56.11	0.0436	0.9174	
	Off-Peak	4942.03	70.3	46.2	0.0487	0.9331	
Transformer-LSTM	Day Peak	5199.23	72.1	67.35	0.0533	0.9096	0.9098
	Evening Peak	6590	81.18	62.76	0.0461	0.9154	
	Off-Peak	13,842.51	117.65	84.64	0.0888	0.9045	
Proposed CNN-LSTM	Day Peak	1075.074	32.7883	25.1459	0.0213	0.9602	0.9515
	Evening Peak	1525.5945	39.0589	29.9743	0.024	0.948	
	Off-Peak	1699.2511	41.222	28.8746	0.0315	0.9464	

Table 5

Forecasting results of CNN-LSTM for 1 through 7 days ahead.

Day Ahead	Peak	MSE	RMSE	MAE	MAPE	R ²	AVG R ²
1	Day Peak	1075.0743	32.7883	25.1459	0.0213	0.9602	0.9515
	Evening Peak	1525.5945	39.0589	29.9743	0.024	0.948	
	Off-Peak	1699.2511	41.222	28.8746	0.0315	0.9464	
2	Day Peak	1921.3073	43.8326	39.043	0.0458	0.8257	0.8492
	Evening Peak	1815.9338	43.7612	33.0553	0.0414	0.8611	
	Off-Peak	1870.6614	43.2848	34.4066	0.0435	0.8508	
3	Day Peak	1914.5774	43.7559	31.8776	0.0405	0.8252	0.8223
	Evening Peak	1989.3789	44.6025	38.5598	0.0452	0.8192	
	Off-Peak	7921.8973	89.0050	38.7895	0.0458	0.7357	
4	Day Peak	7015.9338	83.7612	33.0887	0.0414	0.7611	0.7758
	Evening Peak	1900.4577	43.5942	31.8776	0.0405	0.8308	
	Off-Peak	1914.5774	43.7559	31.8776	0.0405	0.8252	
5	Day Peak	10,921.3073	104.5051	79.043	0.0658	0.5957	0.7292
	Evening Peak	7015.9338	83.7612	63.0553	0.0514	0.7611	
	Off-Peak	5370.6614	73.2848	54.4066	0.0605	0.8308	
6	Day Peak	15,093.6481	122.8562	94.7827	0.0793	0.4426	0.5723
	Evening Peak	12,501.5048	111.8101	87.8776	0.0705	0.5752	
	Off-Peak	9566.3789	97.8079	69.6157	0.0752	0.699	
7	Day Peak	15,477.3321	124.4079	94.6881	0.0772	0.43	0.5594
	Evening Peak	11,555.7476	107.4977	82.3541	0.0648	0.6084	
	Off-Peak	11,470.2638	107.0993	80.1905	0.0857	0.6399	

the exact same model was given 60 % of it to train and showed an excellent result across the different peaks. The models were able to capture the usage pattern of both Dhaka and Sylhet, giving 0.888 and 0.8212 in R², on average across the peaks and off-peak, for the cities, respectively. The day peaks were captured particularly well, as seen in Table 6.

The considering of such specific features, enables better analysis of the load patterns, allowing the model to grasp what is usually ignored in traditional studies and the intricate fluctuations of energy usage of the entire city. The concentration on the sub segments allows the hybrid 1D-CNN-LSTM model to deal with the variance of energy consumption, thus making the forecasting of energy usage variables more than what the current models have been able to achieve. This segmentation also

Table 6

Result of CNN-LSTM for Dhaka and Sylhet.

City	Peak	MSE	RMSE	MAE	MAPE	R ²
Dhaka	Day Peak	1673.531	40.9088	33.0148	0.0182	0.936
	Evening Peak	2644.385	51.4236	39.8564	0.0209	0.8858
	Off-Peak	3651.415	60.4269	49.9276	0.0264	0.8423
Sylhet	Day Peak	2563.0128	50.6262	40.4169	0.022	0.9019
	Evening Peak	6448.7413	80.304	71.0628	0.0374	0.7213
	Off-Peak	3695.1581	60.7878	51.6046	0.0273	0.8403

provides better resolution and detailed understanding of consumption patterns which enables electricity providers to take informed decisions on how they should make efficient use of the available resources for maximum efficiency.

To confirm the effectiveness and robustness of the model by its ability to deal with uncertainties, a controlled robustness analysis was undertaken which involved the introduction of Gaussian noise to the meteorological features. Controlled noise levels of 5 %, 10 %, 15 % and 20 % were added to the predictor variables of Temperature, Dew/Frost Point, Wet Bulb Temperature, Earth Skin Temperature, Min and Max Temperature, Specific and Relative Humidity and Precipitation. The goal of this experiment was to determine the ability of the model to function in realistic, uncertain and fluctuating environments where unpredictable sensor malfunctions and rangebound errors form the basis of fluctuating 'real-time' data. The results of this robustness analysis, across the three different demand intervals (Day Peak, Evening Peak, and Off-Peak), are presented in Table 7.

The verified robustness of the model is commendable, as it holds true for all three demand intervals, while maintaining good alignment to load values, even under extreme noise perturbations. There are still slight discrepancies at extreme load intervals; for instance, during sudden spikes or dips; but the load trend and magnitude is still preserved. More notably, the off-peak performance enhances with the addition of noise, which further boasts the model's ability track and account the discrepancies. More importantly, the narrow and well-contained

Table 7

Performance analysis after adding noise.

Time	Gaussian Noise (σ)	MSE	RMSE	MAE	MAPE	R ²
Day Peak	0 %	1075.074	32.7883	25.1459	0.0213	0.9602
	5 %	2068.996	45.4862	34.1382	0.0291	0.9233
	10 %	2057.024	45.3544	34.579	0.0296	0.9238
	15 %	2281.287	47.7628	36.3314	0.031	0.9155
	20 %	2586.298	50.8556	39.8925	0.0339	0.9042
Evening Peak	0 %	1525.595	39.0589	29.9743	0.024	0.948
	5 %	2113.397	45.9717	33.5935	0.0274	0.928
	10 %	2235.118	47.277	34.4574	0.0283	0.9238
	15 %	2362.169	48.6021	34.9769	0.0284	0.9195
	20 %	2803.865	52.9515	38.7051	0.0316	0.9044
Off-Peak	0 %	1699.251	41.222	28.8746	0.0315	0.9464
	5 %	1686.003	41.061	27.7656	0.0309	0.9469
	10 %	1692.761	41.1432	27.0482	0.0301	0.9466
	15 %	1725.826	41.5431	28.308	0.0316	0.9456
	20 %	1659.113	40.7322	29.5364	0.0329	0.9477

prediction envelopes, indicative of bounded, un-expanded envelopes, suggest that the hybrid CNN-LSTM architecture is able to generalize appropriately, and is not hyper responsive to noisy feature inputs.

Fig. 11 shows the CNN-LSTM model predictions for the Day Peak, Evening Peak and Off-Peak periods of demand periods for the entire range of modeled noise conditions. The plots are drawn with the actual load (blue) and superposed load predictions in the different noise levels. The shaded area shows the range of predictions which spans the entire 0 to 20 % noise range.

This experiment shows that the proposed approach is capable of tracking the inbuilt uncertainties in the meteorological data, and simultaneously, preserving the high accuracy of the forecast. This robustness is essential for the proposed approach as weather-based predictors are, in practice, far from perfect. This is counter parted by the sustained performance under controlled noise injection, which strengthens the technical contribution of the study and further endorses the practicality of CNN-LSTM driven smart urban energy systems.

According to the noise injection test, the hybrid CNN-LSTM model is not only accurate, but also resilient to the uncertainties present in the meteorological inputs. The CNNs learning layers derive stable local patterns associated with the input featured, suppressing structureless noise and other weak signals introduced by disorderly disturbance. These features are sent to LSTM layers, which learn to emphasize the essential trends in an input time series. The value of useful and coherent input time series heavily relies on the level of distortion the LSTM model is able to withstand. This structure ensures a disproportionate amount of short-term noise does not become infused into the LSTM predictions, which is essential to creating coherent and bounded forecast envelopes. To the extent, the model can hold on to the trend and magnitude of the load values despite noisy and imperfect predictors, it can be concluded that data imperfections are a distinguishing mark of operating in the concrete reality of the world. Therefore, the model smartly tackles uncertainty as it integrates CNN features with the LSTM noise-resilience in

sequential time order. The research in this context has established a new energy forecasting paradigm in terms of the suitable approaches towards the design of more insightful forecasting systems for real-time data.

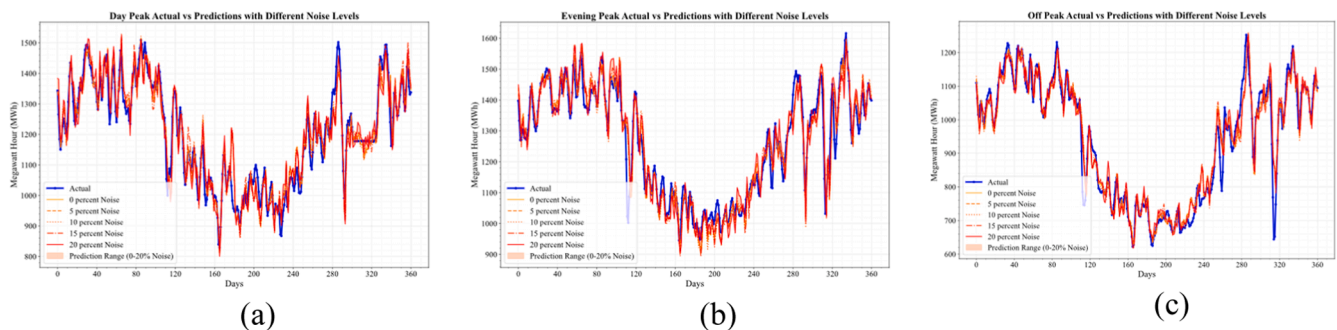
The hardware configuration is very important in the task of electricity load prediction as it relates to the training and inference velocity of the model and determines its ability to run in real-time and to scale. The following configurations were considered in order to compare how the hardware devices can influence the performance and efficiency of the proposed model. Table 8 shows the hardware setup for training the CNN-LSTM model, we compared the two configurations where one uses only CPU and another one uses a GPU. For the CPU-only setup, it contains about 2 (Xeon, 2.2 GHz) vCPUs, 13 GB memory, no GPU and 33–100 GB of disk. The "With T4 GPU" configuration is identical to CPU-only in terms of CPU and RAM, but adds an NVIDIA Tesla T4 GPU with 16 GB of VRAM, with disk storage also being similar. This configuration represents the experimental grounding of the hardware used to train the model and the practical applicability of the hardware.

For the CPU mode, the training range from 10 m 13 s to 10 m 20 s and the inference runs between 343 ms and 345 ms, while time is remarkably compared with the GPU hardware where the training ranges from 2 m 9 s and 2 m 12 s and the inference time goes from 130 ms to 132 ms, while has nearly no variation with the IGA time around 1 m 3 s to 1 m 27 s. These findings clearly indicate that the GPU facilitate faster processing

Table 8

Comparison of both setups in different intervals for the city of Chittagong.

Mode	Peak	Training Time	Inference Time
CPU	Day Peak	10 m 3s	345ms
	Evening	10 m 20s	343ms
	Off	10 m 13s	344ms
GPU	Day Peak	2 m 12s	131ms
	Evening Peak	2 m 10s	132ms
	Off-Peak	2 m 9s	130ms

**Fig. 11.** Actual vs predicted plots with noise for (a) day peak; (b) evening peak and (c) off-peak.

and inference time than CPU. Moreover, the computational performance comparison from the Table 9 clearly outlines a remarkable order-of-magnitude reduction in computational cost with the GPU utilization. The CPU configuration performs training longer than 10 min and is shortened to around 2 min in the GPU one. For the inference time's as well, we see decreasing times which demonstrates the GPU's advantage of the computational load reduction. Following this, Table 9 presents training and inference times for the CNN-LSTM model applied to four months of load data for Dhaka and Sylhet, compared with Table 8 which used a full year of data.

The CPU configuration reveals training durations of nearly 6–7 min, whereas GPU training durations drop to 1.5 min, and the inference times remain relatively stable (321–328 ms for CPU, 123–129 ms for GPU). This shows that the GPU seems to speed up computation even for short time series and sparse data points. More importantly, this indicates that the method scales up effectively: for much larger test cases with thousands of variables, training time would increase linearly, and the method would still retain its GPU advantage, proving its practicality for large-scale optimization problems with sustained inference stability.

The obtained results show the capability of the proposed hybrid CNN-LSTM model for electricity load forecasting and that it performs well with proper lag values and hyperparameter tuning. The model achieves new advances compared to the state-of-art models on several metrics and demonstrates strong predictive abilities for both short-term and days-ahead periods. Furthermore, the hardware setup comparisons show the large improvement in model performance with the introduction of GPU acceleration, resulting in faster training and inference times, which are important for real-time applications within urban energy systems. Our observation confirms the effectiveness and generality for the proposed method to practically address large-scale energy forecasting.

In comparison with the similar works on this domain, the originality of this research lies in the proposing of a new technique using segmented data, which divides the daily energy consumption into the day peak, evening peak and off-peak periods. Despite the variability in data created through this segmentation, the proposed study successfully matched and even outperformed most of the state-of-the-art models. Table 10 compares the related works with the proposed research.

Upon comparing the results of the proposed method, improvements can be noticed, not only in the raw results, but also in the ability to effectively capture the load variability patterns that are present in practical use case within a city. This demonstrates the selection of the 1D-CNN-LSTM model in forecasting energy demands in urban scenarios.

5. eXplainable AI (XAI): IGA insights across peak and off-peak energy forecasting

IGA is a widely adopted attribution-based XAI technique designed to interpret the decision-making process of deep learning models, often considered as "Black Box" due to their complex internal structures. In the context of this research, IGA applied to interpret the CNN-LSTM model's predictions of electrical load by identifying which meteorological and temporal features most significantly influence the forecast.

To enhance the interpretability of the proposed CNN-LSTM model in forecasting electrical load across varying daily demand intervals, IGA

method has been employed which is a robust XAI approach designed to assess the contribution of input features across time. IGA compute the gradients of the model's output with respect to each input feature, averaged over a range of interpolations from a baseline to the actual input. This allows for more nuanced insight into feature importance, especially in deep neural architectures. Figs. 12, 13 and 14 illustrate the attribution heatmaps for Day Peak, Evening Peak and Off-Peak load forecasting, respectively, across seven lag points (t-7 to t-1). Each heatmap captures the average contribution of individual meteorological features such as temperature, humidity parameters and precipitation on the model's prediction.

In the Day Peak heatmap (Fig. 12), the most significant attributions are observed in recent lag steps, particularly t-2 to t-1 for forecasting, where the model heavily relies on specific humidity (114.57, 115.48), dew point (149.13, 57.67) and ambient temperature (54.46, 11.87). This suggests that short-term thermal and moisture conditions are critical indicators for daytime energy demand spikes. Notably, the model's focus intensifies closer to the current time step, reflecting the real-time sensitivity of peak load dynamics.

For the Evening Peak period (Fig. 13), IGA results similarly emphasize the importance of specific humidity (lag t-3: 132.21) and dew point (lag t-3: 111.75), along with a strong attribution to recent total load values (lag t-1: 214.60). This indicates the persistence of humidity-driven consumption behaviour into the evening hours, likely due to sustained cooling demand. Moreover, a greater spread of attribution across mid-range lags (t-4 to t-2) implies that the model integrates broader temporal information during evening periods.

In the Off-Peak interval (Fig. 14), although the total attribution values are slightly lower than peak periods, specific humidity (98.17 at t-3 and 92.07 at t-2) and dew point again emerge as dominant contributors. Interestingly, earth skin temperature and wet bulb temperature also show relevant but moderate contributions, indicating their indirect but meaningful role in base load estimations.

Overall, the IGA analysis reveals that humidity-related variables and thermal features play a consistent and dominant role across all load periods, with more recent lags (particularly t-3 to t-1) being the most influential. These insights validate the model's temporal structure and reinforce the selection of lag-based input windows for effective learning. By integrating XAI through IGA, the proposed model offers not only high predictive accuracy, but also transparency and trust, which are crucial for operational decision-making in dynamic-urban smart grid-based energy systems.

6. Discussions

This study aimed to develop a robust and interpretable deep learning model for short to medium-term energy load forecasting in an urban metropolitan setting by leveraging a hybrid CNN-LSTM architecture. The model was evaluated across multiple daily load intervals such as Day Peak, Evening Peak and Off-Peak, using a range of most popular performance metrics for regression (MSE, RMSE, MAE, MAPE and R^2). In addition, the model's transparency was enhanced through XAI using IGA to quantify the contribution of environmental and temporal features to forecast outcomes.

The proposed hyper-tuned 1D-CNN-LSTM model consistently outperformed baseline models like CNN, LSTM, GRU and Transformer-LSTM in both single-day and multi-day ahead forecasting tasks. It achieved the lowest RMSE (<41.22) and MAPE (<0.031), along with the highest R^2 (0.95) values, particularly on the 1-day ahead forecasts where the model maintained exceptionally high accuracy across all peak periods. Performance gradually declined with increased forecast horizon, which is expected due to accumulating temporal uncertainty. However, the model sustained practical accuracy up to Day 4, making it well-suited for operational energy planning.

A key strength of this work lies in the lag value analysis, where different historical time steps were tested to determine the optimal lag

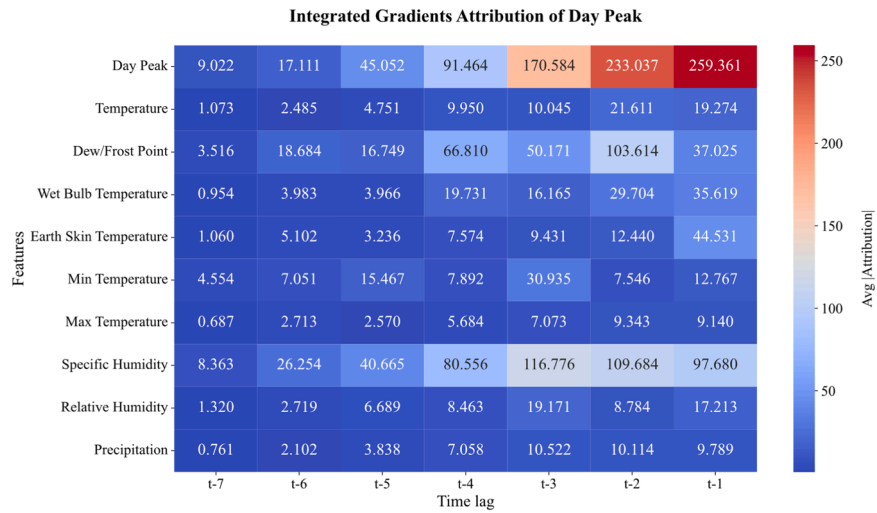
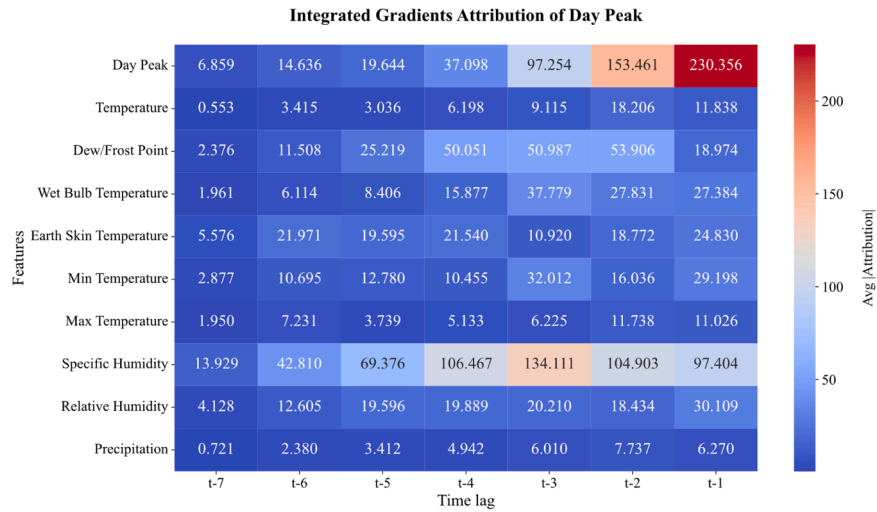
Table 9
Training and inference times for Dhaka and Sylhet dataset.

Mode	Peak	Training Time	Inference Time
CPU	Day Peak	6 m 45s	328ms
	Evening	6 m 58s	321ms
	Off-Peak	6 m 40s	326ms
GPU	Day Peak	1 m 27s	129ms
	Evening Peak	1 m 25s	125ms
	Off-Peak	1 m 26s	123ms

Table 10

Comparison of results with some state-of-the-art related works.

Reference	Year	Proposed Model	RMSE	MAE	MAPE	R ²
[46]	2020	GRU-CNN	1203.23	-	2.88 %	-
[32]	2020	DNN	0.04	0.02	0.21	0.99
[30]	2023	ANN, ARIMA	-	-	173.66, 689.21 %	-
[19]	2023	TPA-LSTM	73.73	51.43	4.41 %	-
[18]	2023	CNN-LSTM + Self-attention	-	-	0.351 (Res), 0.027 (Gen)	-
[48]	2023	CNN	-	-	1.97	-
[29]	2023	GRU	-	0.03266	-	90.23 %
[43]	2024	Seq2Seq-LSTM-Attention	-	0.3027	-	-
[45]	2024	Hybrid LSTM + Online Correction	352.044 (BE), 59.91 (DK), 457.098 (NO)	255.718 (BE), 42.55 (DK), 339.112 (NO)	2.64 % (BE), 2.692 % (DK), 2.152 % (NO)	-
Proposed	2025	CNN-LSTM	37.6897 (avg)	27.9983 (avg)	0.0256 (avg)	0.9515 (avg)

**Fig. 12.** Integrated Gradients Attribution heatmap for energy load forecasting for day peak period.**Fig. 13.** Integrated Gradients Attribution heatmap for energy load forecasting for evening peak period.

structure. A lag window of 7 time-steps was found to offer the best trade-off between input richness and model stability, enabling the network to capture both short-term volatility and temporal patterns in energy demand. This finding was further substantiated by the model's consistent performance across different daily cycles, including the more variable Off-Peak loads.

Moreover, the robustness experiments with the Gaussian noise

indicated that the model could accommodate uncertainties to the predictors, including temperature, humidity and precipitation. CNN layers filtrate high-frequency fluctuations and LSTM units maintain meaningful temporal dependencies, avoid the random disturbance dominating forecast. The model also performed well at extremely noisy cases with a SNR of up to 20 % ($R^2 > 0.90$), demonstrating its robustness to imperfect real-world noise. Such a capability for absorbing uncertainty and to

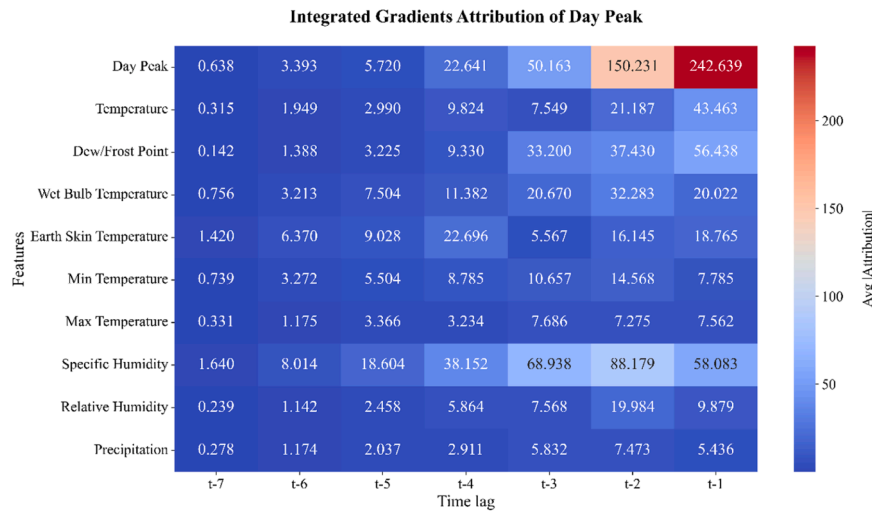


Fig. 14. Integrated Gradients Attribution heatmap for energy load forecasting for off-peak period.

remain stable is of crucial need for smart grid applications for which sensor errors and variable environment inputs are inevitable.

In order to address these uncertainties associated with the real-world data, robustness testing was conducted by adding Gaussian noise (5–20 %) to the meteorological inputs. The framework with CNN-LSTMs displayed stability with the perturbations, with an average deviation of 3.0 % for the Day Peak interval, 2.1 % for Evening Peak and 0.6 % for Off-Peak intervals, with an overall average deviation of 2.5 % over all three periods. This demonstrates the ability of the model to cope with noisy or imperfect inputs, and so enhances confidence in its application to real-world practical forecasting. It has also been validated that the model is capable of scaling up to larger smart grid systems.

Again, to address the interpretability challenge posed by deep learning's "Black-Box" nature, Integrated Gradients were used to reveal the internal reasoning behind the CNN-LSTM model's predictions. The attribution heatmaps across Day Peak, Evening Peak and Off-Peak intervals demonstrated a consistent reliance on humidity-related features (particularly specific humidity and dew point) and short-term lags (t-3 to t-1). These patterns were strongly aligned with the earlier correlation analysis, thereby validating the physical and meteorological relevance of the selected features. Such interpretability adds an important layer of model transparency, essential for real-world adoption in critical infrastructure settings like energy grids.

Moreover, the comparative IGA results highlight that while Day Peak forecasts are more sensitive to immediate meteorological changes, Evening Peak and Off-Peak intervals integrate a slightly wider temporal context. These distinctions could inform the design of future adaptive or period-specific forecasting systems.

Overall, the proposed CNN-LSTM with IGA-based XAI not only enhances forecast accuracy but also delivers actionable interpretability, positioning it as a promising solution for smart grid operations, demand response strategies and urban energy resilience planning. The integration of deep learning and explainability provides a strong foundation for advancing data-driven decision-making in climate-sensitive urban environments.

7. Conclusion

In this study, a hybrid CNN-LSTM deep learning framework was proposed for short-term energy load forecasting across distinct daily demand intervals such as Day Peak, Evening Peak and Off-Peak in an urban environment. The model demonstrated superior performance compared to conventional architectures (CNN, LSTM, GRU and Transformer-LSTM), achieving high prediction accuracy and

generalization capability, particularly in 1-day ahead forecasts with an average R^2 exceeding 0.95. The incorporation of lag value analysis further optimized temporal input selection, with a lag window of 7 proving most effective for capturing historical dependencies in load patterns.

To overcome the interpretability limitations of black-box models, the IGA method has been employed, enabling clear attribution of feature importance across multiple time lags. The IGA analysis revealed that specific humidity, dew/frost point and temperature-related parameters were the most influential predictors across all time intervals, with recent lags (t-3 to t-1) playing a critical role in the model's decision-making process. These findings are consistent with domain knowledge and prior correlation analysis, enhancing confidence in the model's internal logic. The model also exhibited remarkable robustness toward input uncertainties, yielding sensible predictions in the presence of noise and imprecise data. The robustness test with Gaussian noise applied to the meteorological inputs confirmed that the CNN-LSTM framework can withstand certain input variability while still reliably forecasting the Day Peak, Evening Peak and Off-Peak periods. This robustness guarantees sound urban energy control decisions and further supports the model's realistic use for smart grid functions.

The combination of high forecast accuracy, AI explainability and temporal flexibility with three distinctive day-evening peaks and off-peak demand intervals make the proposed 1D-CNN-LSTM architecture a strong candidate for integration into real-time energy management systems, demand-side planning and urban smart grid optimization strategies. By bridging the gap between predictive performance and interpretability, this research contributes to the development of more transparent, intelligent and actionable forecasting tools for future smart city applications.

8. Limitations and future work

Although the proposed CNN-LSTM model achieved strong forecasting accuracy and interpretability, several limitations remain. Forecast performance declined beyond 3 to 4 days ahead, indicating the need for improved long-term temporal modelling. Additionally, the model did not account for external factors like holidays or socio-economic events, which could enhance prediction robustness.

Future research should explore generalizing the model across different climatic zones with much bigger dataset. Besides, privacy aware Federated Learning integration could be a promising option for future scope too. Lastly, optimizing the model for real-life deployment will require work on computational efficiency and system integration.

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CRediT authorship contribution statement

Sarowar Morshed Shawon: Writing – review & editing, Supervision, Formal analysis, Data curation, Conceptualization. **Shah Nawaz Haider:** Visualization, Validation, Investigation, Formal analysis. **Arnab Barua:** Writing – original draft, Methodology, Formal analysis. **Steve Austin:** Writing – original draft, Formal analysis. **Ifaz Ahmed Adan:** Writing – review & editing, Formal analysis. **Mohammad Shahadat Hossain:** Writing – review & editing, Supervision. **H.T. Zubair:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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